

An Interdisciplinary Approach to Combating Social Engineering Cyber Threats

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IDS 300W: Interdisciplinary Theory and Concepts

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December 5, 2024

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Abstract

Social engineering cyberattacks take advantage of vulnerabilities in technical systems, human behavior, and legal gaps, which present substantial risks to individuals and organizations. This paper argues that this complicated threat necessitates a comprehensive, interdisciplinary approach that combines observations from computer science, sociology, psychology, and law. Sociological perspectives emphasize the organizational norms and social dynamics, including authority and trust, which increase an individual's risk of being victimized. At the same time, psychology explores emotional and cognitive weaknesses, such as heuristics and biases, that attackers are known to target specifically. Computer science offers technical solutions, including artificial intelligence (AI) and machine learning (ML), which can be implemented into cybersecurity systems to recognize and resolve threats, and legal frameworks require organizational accountability and encourage international cooperation. Combining each perspective, this paper suggests that policymakers and organizations create adaptive defenses, increase awareness, and enhance regulations. This interdisciplinary approach highlights the significance of collaborating to offset the growing threat of social engineering cyberattacks.

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Introduction

Social engineering cyberattacks exploit vulnerabilities in technical systems, human behavior, and legal gaps, presenting significant challenges to individuals, organizations, and governments. Such attacks employ a sense of trust, the appearance of an authority figure, or invoke an emotional response to fulfill malicious desires, including initiating data breaches or successfully committing financial theft. Many organizations have not successfully created defense mechanisms to keep up with these growing threats. Social engineering cyberattacks are successful because they are designed to exploit systemic and human vulnerabilities, but organizations can overcome such threats by implementing a flexible defensive strategy that focuses on both human and technological factors.

Discussion of the Interdisciplinary Research Method

Focusing on the complicated issue of social engineering cyberattacks necessitates an interdisciplinary approach because their success can be attributed to the vulnerabilities of more than one system. These attacks thrive from the weaknesses of technical systems, human behavior, and legal frameworks. Therefore, no single discipline can thoroughly address the varying nature of this type of threat. Computer science is relevant to the problem because it can provide the methodologies and tools necessary to identify and mitigate cyberattacks, and sociology explores the organizational norms and other social structures that affect an individual's risk. Psychology offers insights into emotional and cognitive vulnerabilities, while law creates accountability and encourages international cooperative efforts. This paper aims to create a detailed understanding of social engineering and offer strategies that collectively focus on both technical and human vulnerabilities.

Integrated Literature Review

Theme 1: Exploitation of Human Behavior

Sociology: Trust and Organizational Norms

Social engineering cyberattacks typically exploit organizational norms and invoke a sense of trust, which are relevant concepts within a sociological context. Trust has a crucial role in social interactions, and attackers leverage trust to trick individuals into exposing confidential information or other harmful choices. Gururaj, Janhavi, and Ambika (2024) highlight that social engineering techniques work because they exploit common patterns of trust, which creates an opportunity to deceive the victim (p. 3). Furthermore, social engineering attacks can occur through various methods, including, but not limited to, “pretexting, phishing, baiting, tailgating, and impersonation” (Gururaj et al., 2024, p. 23). While this makes social engineering seem complex, each attack commonly utilizes a few critical strategies across all the previously mentioned attack vectors, including “impersonation, emotional manipulation, and the exploitation of trust, authority, or fear” (Gururaj et al., 2024, p. 23). This allows social engineering cyberattacks over text, email, and social media networks because regardless of the platform chosen (depending on which is accessible to a chosen victim), each allows for a socio-psychological technique to be utilized for manipulation. Interestingly, while Gururaj et al. discuss socio-psychological techniques in detail, they also propose the concept of including machine learning (ML) and artificial intelligence (AI) as components of an organization’s security policy. This is due to how these technologies can be used to develop an application

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capable of neural language processing (NLP). This defensive technique would be sufficient for identifying “certain phrases and sentence patterns” that the system would warn users about before acting (Gururaj et al., 2024, p. 36). Email service providers commonly use NLP to identify, segregate, and mark phishing emails for deletion, but developers could be prompted to work towards expanding this implementation to other platforms.

Workplace dynamics and organizational norms, such as hierarchies, can inevitably increase an individual’s susceptibility to social engineering by failing to encourage the habit of exercising critical thinking skills. Wulandari, Adnan, and Wicaksono (2022) indicate that social engineering attacks succeed by exploiting “trust, motives, habits, and behavior of individuals to manipulate them” (Introduction). For example, Wulandri et al. (2022) stated that cyberattacks “increased 600% during the [COVID-19] pandemic,” which can be attributed to teleworking, necessitating a high dependency on the Internet (Introduction). This frequently occurring exposure was something that many individuals were not prepared for, as it requires an organization to adequately train its employees on how to review communications to determine whether their requests are legitimate and ethical. The long-lasting fear and anxiety many felt throughout the pandemic “actually drove the success of cyberattack incidence,” resulting in a higher occurrence of compliance with malicious requests (Wulandari et al., 2022, Introduction). This is further worsened by a common tendency for victims to either defer to their organization’s authority figures or provide habitual responses without thoroughly considering whether the action should be taken. As Wulandari et al. (2022) highlight, “humans naturally tend to trust one another compared to computers, making them a soft target for this approach” (2.1 Literature Review). While employees may commonly feel pressured to remain compliant instead of

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questioning directives, workplaces must create a culture that encourages critical thinking and questioning regardless of how they perceive the sender's authority.

Psychology: Cognitive Biases and Emotional Triggers

Psychology deeply investigates specific emotional responses and thought processes that social engineers aim to exploit. Cognitive biases, including confirmation bias and overconfidence, frequently result in individuals making poor decisions while under pressure. As Longtchi, Rodriguez, Al-Shawaf, Atyabi, and Xu (2024) illustrate, "Humans are susceptible to SE attacks partly because they use shortcuts in reasoning and are often overconfident and absentminded" (p. 221). Attackers frequently take advantage of such biases by creating a scenario that accurately aligns with a victim's typical expectations, which decreases the likelihood of each victim becoming suspicious of the request's origin. For example, urgency bias is a frequent target; as Longtchi et al. (2024) described, "Urgency ... a situation which requires immediate action" causes "a decrease of chance in recognizing an attack" (pp. 223-224). This manipulation strategy is knowingly employed because "SE [social engineering] attackers exploit some PFs [psychological factors] to encourage individuals to comply with their SE attacks" (Longtchi et al., 2024, p. 214).

Emotional triggers, including trust, fear, empathy, authority bias, and social conformity, pertain to various other integral components of psychological analysis, which help demonstrate social engineering cyberattacks' success. Longtchi et al. (2024) expand on this by describing how "situations that evoke fear incur a strong avoidance reaction in both behavioral responses and cognitive processing" (p. 221). Likewise, attackers frequently use "empathy as an intuitive

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behavior or a persuasion technique to achieve their goals” (Longtchi et al., 2024, p. 222). When an attacker succeeds in this manner, they can convince the victim to complete a specific action, including opening a malicious link or even sending money. For instance, a phishing email could consist of an emotionally charged message to decrease the likelihood of an individual using logical reasoning, resulting in the victim complying with the message’s request. Awareness training programs are fundamental to resisting such tactics; they train individuals on these psychological principles to identify these manipulative techniques.

Theme 2: Technical Solutions and Their Limitations

Computer Science: Cognitive Accessibility and Agile Cybersecurity Frameworks

The computer science discipline can be examined to observe how organizations can supply their staff with the necessary technical resources to defend against social engineering cyberattacks. For example, organizations should strongly consider implementing an agile cybersecurity framework. This is inspired by the Agile software development lifecycle, which depends on continuous improvement and collaborative efforts among developers and security experts. Therefore, it is naturally an adaptive solution by serving as a guide for security experts to install and configure the necessary software patches, train users, analyze data flow, complete various analyses, verify accurate functionality of numerous services, and provide support to relevant users and other staff within the organization. In their study, computer science researchers from Indonesia determined the necessary components of a rebuilt cybersecurity framework by using the Q-methodology, which allowed them to present their survey participants with a selection of statements related to cybersecurity concerns that participants can choose

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from. By identifying specific factors that were commonly shared, Handri, Sensuse, and Tarigan (2024) “revealed five core components: Security Governance, Risk Management, Incident Management, Security Technology, and Organizational Culture” (p. 108835). While exploring each criterion of each component is extensive and potentially expands beyond the scope of this paper, this alone demonstrates how the focus of an agile method is not limited to using complex technologies; it also “includes operational flexibility and organizational response to security changes” (Handri et al., 2024, p. 108837). Furthermore, agile frameworks are intended to regularly repeat each process across all supported technical systems to ensure that security measures are continually improved. The researchers described this characteristic after they discussed the creation of a cybersecurity framework to address the insights gathered by survey participants, as “regular and thorough security testing at each iteration also improved the overall quality and security of the framework” (Handri et al., 2024, p. 108846). By integrating each detailed core component, agile cybersecurity frameworks can offer an extensive focus on social engineering cyberattacks by combining technological improvements with an organization’s ability to adapt to new threats as they emerge in the long-term.

Although technical solutions are critical in dealing with cyberattacks, they cannot protect against human vulnerabilities alone. Naqvi, Kävrestad, and Islam (2024) emphasize how cognitive factors, including the inability to remember a complex password, can make systems either inaccessible or otherwise risky to use: “Users with cognitive disabilities do have difficulties while authenticating using the current range of authentication options available; for instance, remembering complex passwords without writing them down can prove to be a cumbersome task” (p. 74). Additionally, for users who do not have a disability, the mental

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workload associated with numerous security functions can also result in a lack of usability, as “minimizing cognitive workload is an important usability factor for cybersecurity functions and interfaces” (Naqvi et al., 2024, p. 74). This indicates the significance of designing systems that satisfy security requirements and user-friendliness. Moreover, accessibility problems affect more than just individuals; they impact more comprehensive societal functions. This relates to how Naqvi et al. (2024) warn, “nonconsideration of the cognitive accessibility in the design and development of cybersecurity functions can impact people’s access to, for instance, online education facilities, health-care services, Internet banking, etc., all of which require interacting and managing cybersecurity functions” (p. 74). Focusing on such restrictions necessitates a human-focused approach to designing cybersecurity systems, which involves streamlining user interfaces (UIs) and integrating automation to increase the accessibility of technical resources for diverse groups.

Sociology: Accessibility and Workplace Hierarchies

Sociology pairs well with the technical approach of computer science by focusing on organizational and other societal factors that affect the adoption and success of cybersecurity efforts. Accessibility is a top priority because variations in digital literacy, combined with the availability of relevant technical resources, can compromise the effectiveness of cybersecurity solutions. Murero (2023) emphasizes how “CIB [coordinated inauthentic behavior] campaigns mislead others by acting with pre-arranged exceptional similarity and ‘secret’ operations,” meaning that attackers often attempt to target individuals whom they perceive as being unaware and unprepared (p. 1). These malicious efforts commonly take advantage of vulnerable groups, as Murero (2023) describes them as “hard-to-detect CIBs’ manipulative operations” that “could

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be one of the greatest threats to freedom of expression and democracy” (p. 2). These modern social engineering strategies, especially during notable events like the vaccination campaign during the COVID-19 pandemic, threaten those affected by a gap in digital literacy and unequal exposure to cybersecurity education. This example refers to Murero’s CIB research, which focused on the incident involving an anti-vaccine adversarial network (AN) finally taken offline by Meta as recently as 2021. The actors in this group used “real, fake, and duplicated accounts” to oppose “the healthcare dictatorship” by using “multiple digital platforms to psychologically mass-harass specific targets” (Murero, 2023, pp. 3-4). To avoid being attacked online or otherwise persuaded to agree with potentially dangerous opinions, disparities in digital literacy and cybersecurity knowledge need to be addressed in relevant organizations by providing their members with training programs and updating their organizational policies to reflect these cybersecurity changes.

Workplace hierarchies play a crucial role in affecting the adoption of cybersecurity solutions. Organizations that establish strict hierarchical structures tend to intimidate employees by making them uncomfortable verifying instructions regardless of whether they seem questionable. This structure can also hinder cross-hierarchical communication efforts, interfering with the spread of significant training opportunities and cybersecurity protocols. This relates to a point made by one of the sociology sources I referenced in the previous sociology section within *Theme 1* when Wulandari, Adnan, and Wicaksono (2022) asserted that “a lack of understanding about security for interacting in cyberspace, combined with limited access to digital resources, exacerbates existing disparities and leaves vulnerable groups particularly susceptible to social engineering” (Abstract). Cybersecurity needs to progress to focus on disparities in the

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availability of technical resources and education, especially in geographic areas that do not have sufficient infrastructure and awareness of best practices to decrease cybersecurity risks.

Sociological insights should be integrated into technical systems to effectively focus on the gap between human elements and technological capabilities.

Theme 3: Legal Frameworks for Accountability

Law: Regulatory Standards and Deterrence

A legal perspective must be analyzed to establish accountability by considering relevant regulations, such as the General Data Protection Regulation (GDPR). This regulation requires organizations to implement robust cybersecurity measures and handle incident reporting effectively. For instance, the GDPR “requires data controllers and processors to verify compliance with its standards through specific documentation, such as applicable logs, rules, and procedures” (Rocchi, 2022, Chapter 1: ISO27001 - Definitions and Security Concepts). It is worth noting that while various legislation will continue to emerge over time, the GDPR pertains to “roughly 400 million people in 28 countries,” which enhances its effectiveness by being enforced worldwide instead of being geographically limited (Rocchi, 2022, Chapter 1: ISO27001 - Definitions and Security Concepts). If a national data protection authority (DPA) concludes that a violation has occurred within an organization, they could potentially issue a fine up to “€20 million or 4% of an entity’s worldwide annual turnover, whichever is greater” (Rocchi, 2022, Chapter 3: Data Protection). This substantial fine is intended to be an effective deterrent against non-compliance, which aims to incentivize organizations to focus heavily on continually improving and maintaining cybersecurity measures. This example of a regulatory standard

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ensures that cybersecurity will remain a fundamental part of an organization's administration rather than an insignificant concern.

Law: Operational Challenges in Global Cybersecurity Governance

The operational challenges in maintaining cybersecurity governance worldwide are complex; these challenges arise from the everlasting need to focus on new threats, adhere to various regulatory frameworks, and address technological and human vulnerabilities. Walter Rocchi, an ISO 27001 lead implementer, has described this “proliferation of increasingly complex, sophisticated, and global threats to this information and its systems” as something that should prompt “organizations to take a more integrated approach to information security” (Rocchi, 2022, Chapter 2: Mandatory Requirements). ISO/IEC 27002 frameworks are designed to provide “information security best practices and implementation guidance, which allows ISO/IEC 27001-compliant organizations to defend their information assets effectively by enhancing their risk management operations” (Rocchi, 2022, Chapter 1: ISO27001 - Definitions and Security Concepts). While these technical standards provide detailed guidance to protect sensitive information, corporate governance also has a role in guaranteeing that “businesses have appropriate decision-making processes and controls in place so that the interests of all stakeholders are balanced.” Unfortunately, a significant operational difficulty is commonplace: “A human being is always the weakest link of the security chain” (Rocchi, 2022, Chapter 1: ISO27001 - Definitions and Security Concepts). This compilation of operational concerns from a regulatory perspective highlights the significance of implementing universal cybersecurity standards, coordinating governance practices, and continuously prioritizing training opportunities to decrease human vulnerabilities.

Applications and Syntheses

Integrated Strategy

To effectively counteract social engineering cyberattacks, a strategy must be developed that integrates sociological and psychological insights to focus on human vulnerabilities. Sociology can be explored to understand the role of authority, trust, and workplace dynamics within an organization and how attackers can exploit them. Psychology pairs well with this by observing how cognitive biases and emotional triggers can be abused to manipulate people. Both disciplines affect the creation of training programs that educate employees to identify suspicious activities and think critically about how to react to them. For example, integrating situations that incorporate these sociological and psychological insights within the modules of an educational program can empower employees by teaching them how to recognize phishing schemes and refuse to give in to their manipulative techniques.

Technical solutions and legal frameworks must also be incorporated to create strong defenses. Computer science provides technical resources such as artificial intelligence (AI) and machine learning (ML) to identify and block attacks, even though such resources must also satisfy regulatory standards to guarantee compliance. Legal processes, including the General Data Protection Regulation (GDPR), offer a framework to ensure accountability by requiring organizations to manage satisfactory cybersecurity defenses consistently. Organizations can develop adaptive systems that satisfy compliance requirements and hinder attack efforts by combining technical solutions with legal mandates. This interdisciplinary strategy

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simultaneously focuses on technical, behavioral, and legal concerns to develop an extensive approach to cybersecurity.

Real-World Examples

A real-world example of these interdisciplinary strategies in operation is the implementation of training programs in a workplace to educate employees on how to focus on phishing schemes. Many organizations have begun combining behavioral insights with technical resources to improve their defenses. For instance, phishing attack simulations coupled with training modules that consist of relevant sociological (organizational norms and social dynamics), and psychological (emotional and cognitive weaknesses) insights can assist employees in identifying manipulation tactics and practicing how to respond to them within a safe, sandboxed environment. Such programs are typically designed to track each employee's progress and collect statistics on which concepts are commonly mastered, and which have frequent gaps in understanding, establishing a feedback loop to reinforce learning. This example of combining behavioral and technical strategies demonstrates the benefit of applying an interdisciplinary approach.

Policy recommendations also emphasize the significance of integrating disciplinary insights. For example, organizations and governments can create policies that require consistent employee cybersecurity training. Furthermore, international treaties like the Budapest Convention could be updated to elaborate on more specific challenges related to social engineering. Policies must also encourage accessibility to guarantee that cybersecurity resources and training materials are inclusive among all participants, regardless of their level of technical

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expertise. These interdisciplinary policy proposals illustrate how even collaboration among dissimilar disciplines can lead to developing a thorough and sustainable solution to the growing threat of social engineering cyberattacks.

Conclusion and Recommendations

Social engineering cyberattacks exploit vulnerabilities in technical systems, human behavior, and legal gaps, presenting significant challenges to individuals, organizations, and governments. This paper has shown that integrating computer science, sociology, psychology, and law insights provides a comprehensive foundation for mitigating these threats. Sociology and psychology provide unique perspectives regarding human behavior, which can influence the development of training programs and other methods of raising awareness. Computer science offers technical solutions to recognize and mitigate attacks. Law creates accountability and encourages international cooperative efforts to guarantee that defenses are dependable and enforceable. Looking forward, interdisciplinary research should be applied to seek new intersections between various technical and behavioral disciplines to create fresh solutions to social engineering and other threats that frequently occur within cyberspace. Policymakers need to reinforce international agreements and enforce inclusive cybersecurity policies. Organizations need to consider requiring training programs that are adaptive to their users and the growing complexity of contemporary cyber threats. Stakeholders can develop exhaustive strategies to remediate the dangers of social engineering.

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Extra Credit Opportunity – My Reflection on My Research & Writing Processes

If I could redraft this paper, I would begin by taking a different approach to the research process. I will admit that in the preliminary stages of my work in this course, I lacked a clear understanding of how I would strategically use each discipline to contribute to my paper, which resulted in me exploring a wide range of academic sources without specific objectives for each discipline. In retrospect, I also could have benefited from starting on my outline to enhance my understanding of what I would use each discipline to discuss; this led to me creating my outline immediately before my first rough draft. Furthermore, I determined that my initial draft lacked some fundamental sections, including the Abstract page and the interdisciplinary research method discussion. With all these challenges, I determined a second rough draft was necessary before sharing my work with my peer review partner, Leslie Zeng.

Regarding my research successes, I recognized some interdisciplinary connections between disciplines; one instance occurred between computer science and sociology while discussing cybersecurity frameworks. Also, while deciding how to use each discipline to support my arguments was challenging, I was still satisfied with the sources I selected for each. I spent considerable time browsing the university library's databases and refining my search criteria.

As previously mentioned, it was not easy to decide how I wanted to design my outline to integrate each discipline effectively during my writing process. However, I will note that I was satisfied with my peer review experience with Leslie. She had a few ideas for how I could be more specific with some points I tried to make, which led to rephrasing some content from my second rough draft and including direct quotes to strengthen some of these points.

This culminating research project has reminded me that an iterative process is necessary because the research process improved my writing, while my writing process resulted in me

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determining that I had missing or inconsistent information. Seeing what I was either missing or ineffectively including in my paper allowed me to continuously determine how to address all the points I wanted to make while avoiding excessive redundant information. In future work, I would like to use what I learned from this experience to develop a systematic approach to establishing clear expectations of each discipline and the academic sources associated with each and thoroughly determining the objectives I want to fulfill to create a more informed outline earlier than I inevitably did in this course.